

Wearable Robotics

L13 - The Role of Parallel Elasticity

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Energy consumption is a convex quadratic function of series spring compliance

$$E_m = \int_{t_0}^{t_f} v_{in} i_m dt$$

(Assuming periodicity)

$$= \int_{t_0}^{t_f} \left[\left(\frac{a_1^2}{k_m^2} + a_2^2 b_m \right) \alpha^2 + \left(\frac{z a_1 b_1}{k_m^2} + z a_2 b_2 b_m \right) \alpha + \left(\frac{b_1^2}{k_m^2} + b_2^2 b_m - z \dot{q}_0 \right) \right] dt$$

$$= a \alpha^2 + b \alpha + c$$

where

$$a = \int_{t_0}^{t_f} \left(\frac{a_1^2}{k_m^2} + a_2^2 b_m \right) dt$$

$$b = \int_{t_0}^{t_f} \left(\frac{z a_1 b_1}{k_m^2} + z a_2 b_2 b_m \right) dt$$

$$c = \int_{t_0}^{t_f} \left(\frac{b_1^2}{k_m^2} + b_2^2 b_m - z \dot{q}_0 \right) dt$$

Series Elastic Actuator (SEA)

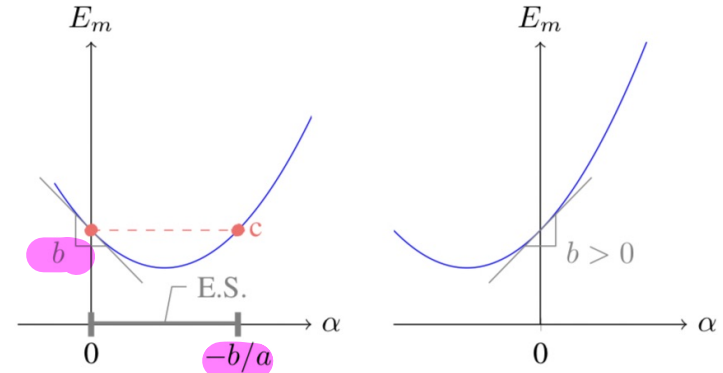
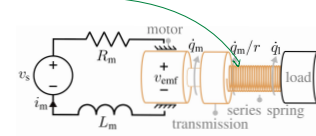


Fig. 2. Left: Energy consumption as a function of compliance, α , where the energy savings (E.S.) region $0 \leq \alpha \leq -b/a$ provides E_m below the rigid level c . Right: Case of motor and load that would not benefit energetically from series elasticity.

* Note: a_1 and a_2 are known constants that depend on the load kinematics and kinetics.

Energy consumption is a convex quadratic function of series spring compliance

$$E_m = a\alpha^2 + b\alpha + c$$

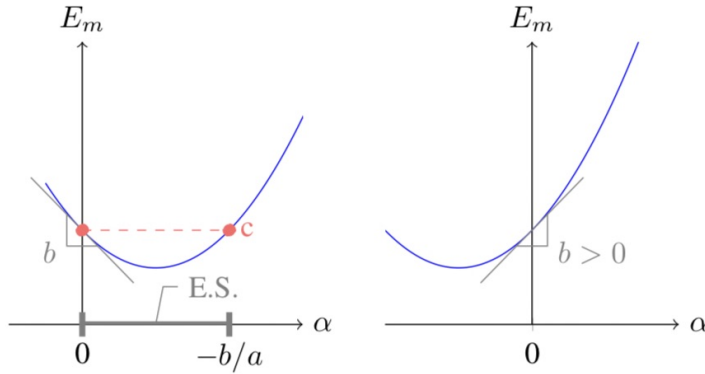


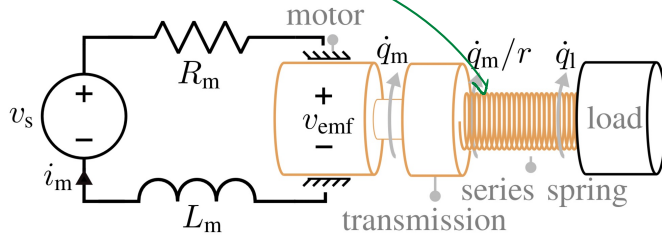
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Properties of this convex quadratic function:

- 1) $\frac{d^2 E_m}{d\alpha^2} = 2a \geq 0$; the function is convex
- 2) $\lim_{\alpha \rightarrow 0} E_m = c$; c is the value of energy required to perform the task without spring
- 3) The value of compliance between 0 and $-b/a$ is the energy savings region
- 4) The value of b defines if it is useful to have series compliance

Takehomes

Series Elastic Actuator (SEA)



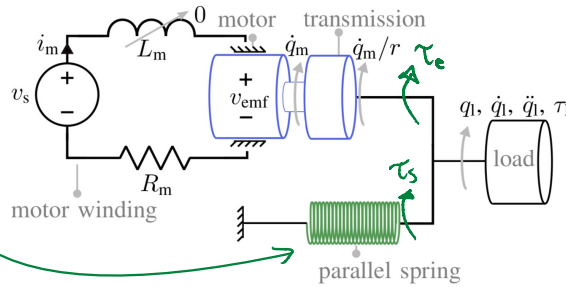
SEAs can only reduce energy losses in periodic motion and when the transmission is fully backdrivable

SEAs decouple load and motor kinematics the series spring can only modify inertial and viscous friction torques

→ Those torques (heat losses) may not be relevant compared to the load torques (DD/QDD)

Modern Direct/Quasi-Direct drive actuators may be able to control torque directly (dynamics of the transmission are negligible). The use of springs in series with DD/QDD is not evident and depends on the application

The diagram illustrates a mechanical system with an electrical motor. The motor winding is connected to a voltage source v_s and has inductance L_m and resistance R_m . The motor's back EMF is v_{emf} . The motor is coupled to a transmission with gear ratio r . The transmission is connected to a load with torque τ_l and a parallel spring with stiffness k . Green arrows indicate the flow of torque and displacement.



Objectives of a parallel spring

Modify Motor Torque

RMS or "Typical" torque

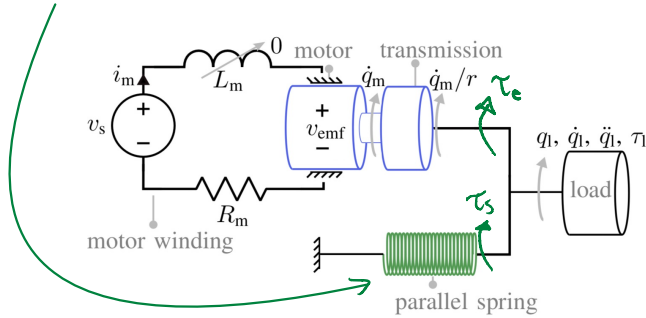
Peak torque

Can you modify
motor kinematics?

Modify energy consumption through the reduction of torque (heat losses)

Effects of parallel springs on RMS motor torque

Parallel Elastic Actuator (PEA)



Dynamic equations:

$$\tau_e + \tau_s + \tau_l = 0 \Rightarrow \tau_e = -\tau_l - \tau_s$$

External torque
on the motor

Torque
Parallel
Spring

Torque load

Parallel spring:

$$\tau_s = -K(q_l + \delta_p)$$

$$= -Kq_l + \tau_{pp}$$

Parallel Spring Preload

K: parallel spring
stiffness

δ_p : spring preload

$$\tau_m = I_m \ddot{q}_m + b_m \dot{q}_m - \frac{\tau_e}{r}$$

$$= I_m \ddot{q}_m + b_m \dot{q}_m + \frac{\tau_l + \tau_s}{r}$$

Note that $q_m = q_l$. Thus,

$$\tau_m = \underbrace{I_m \ddot{q}_l + b_m \dot{q}_l}_{b} + \frac{\tau_l}{r} - \frac{K}{r} q_l + \frac{\tau_{pp}}{r}$$

$$a = \begin{bmatrix} -\frac{K}{r} \\ \frac{1}{r} \end{bmatrix}$$

$$\tau_m = a^T x + b;$$

where

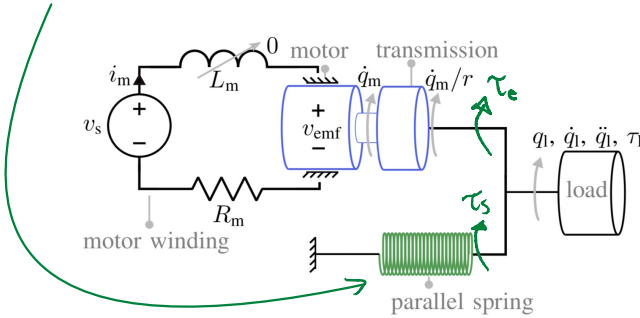
$$x = \begin{bmatrix} q_l \\ \tau_{pp} \end{bmatrix}$$

RMS Torque

$$\text{RMS}(\tau_m) = \sqrt{\frac{1}{t_f - t_0} \int_{t_0}^{t_f} \tau_m^2 dt}$$

Effects of parallel springs on RMS motor torque

Parallel Elastic Actuator (PEA)



$$\tau_m = a^T x + b; \quad \text{where } x = \begin{bmatrix} \kappa \\ \tau_{pp} \end{bmatrix}.$$

RMS Torque

$$\text{RMS}(\tau_m) = \sqrt{\frac{1}{t_f - t_0} \int_{t_0}^{t_f} \tau_m^2 dt}$$

$$\begin{aligned} \tau_m^2 &= (a^T x + b)^T (a^T x + b) \\ &= (x^T a + b^T) (a^T x + b) \\ &= x^T a a^T x + x^T a b + b^T a^T x + b^T b \\ &= x^T a a^T x + 2b^T a^T x + b^T b \end{aligned}$$

From its definition $a \in \mathbb{R}^{2 \times 1}$, thus $a a^T \in \mathbb{R}^{2 \times 2}$

Note that $a a^T$ is a **positive semi-definite matrix** / Open informative tangent

* Definition: the matrix $A \in \mathbb{R}^{n \times n}$ is positive semi-definite if it satisfies

$$z^T A z \geq 0, \quad \text{for any } z \in \mathbb{R}^n$$

For example:

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \text{ is not positive semi-definite}, \quad A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \text{ is positive semi-definite,}$$

$$z = \begin{bmatrix} -1 \\ 0 \end{bmatrix} \Rightarrow z^T A z = \begin{bmatrix} -1 & 0 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \end{bmatrix} = -1$$

why?

A matrix $A \in \mathbb{R}^{n \times n}$ is positive semi-definite if and only if its eigenvalues are positive or zero

In our case:

$$a = \begin{bmatrix} -\frac{q_e}{r} \\ \frac{1}{r} \end{bmatrix}; \text{ then}$$

$$a a^T = \begin{bmatrix} \frac{q_e^2}{r^2} & -\frac{q_e}{r^2} \\ -\frac{q_e}{r^2} & \frac{1}{r^2} \end{bmatrix}$$

The eigenvalues of $a a^T$ are the roots of the characteristic polynomial

$$\begin{aligned} \det(\lambda I - a a^T) &= \det \left(\begin{bmatrix} \lambda - \frac{q_e^2}{r^2} & \frac{q_e}{r^2} \\ \frac{q_e}{r^2} & \lambda - \frac{1}{r^2} \end{bmatrix} \right) \\ &= \lambda^2 - \lambda \left(\frac{1}{r^2} + \frac{q_e^2}{r^2} \right) + \frac{q_e^2}{r^4} - \frac{q_e^2}{r^4} \\ &= \lambda \left(\lambda - \left(\frac{1}{r^2} + \frac{q_e^2}{r^2} \right) \right) \end{aligned}$$

$$\left. \begin{aligned} \lambda_1 &= 0 \\ \lambda_2 &= \frac{1}{r^2} + \frac{q_e^2}{r^2} \end{aligned} \right\} \text{ Eigenvalues are non-negative}$$

They are positive or zero.

Note: You don't need to calculate the eigenvalues explicitly to show positive semi-definiteness.

For example:

A.K.A. Gramian of $B \in \mathbb{R}^{m \times n}$

Any matrix of the form $B^T B$ is positive semi-definite. why?

We can use the singular value decomposition

$$B = U \Sigma V^T$$

diagonal matrix with positive or zero singular values

$$U^T U = I$$

$$V^T V = I$$

$$\begin{aligned} \text{Thus; } B^T B &= V \Sigma^T U^T U \Sigma V^T \\ &= V \Sigma^2 V^T \end{aligned}$$

Matrix of eigenvalues. All of them positive or zero.

/ Close informative tangent

Effect of parallel stiffness and preload torque on RMS torque

$$\tau_m^2 = x^T a a^T x + z b^T a^T x + b^T b$$

$$\text{RMS}(\tau_m) = \sqrt{\frac{1}{T_f - T_0} \int_{T_0}^{T_f} \tau_m^2 dt}$$

Note that the square-root and factor $\frac{1}{T_f - T_0}$ only modify the magnitude of $\text{RMS}(\tau_m)$

However, they do not modify the effect of parallel stiffness and preload

The value of $x = \begin{bmatrix} K \\ \tau_p \end{bmatrix}$ that minimizes

$\text{RMS}(\tau_m)$ is the same value that minimizes

$$\int_{T_0}^{T_f} \tau_m^2 dt$$

$$\int_{T_0}^{T_f} \tau_m^2 dt = \int_{T_0}^{T_f} (x^T a a^T x + z b^T a^T x + b^T b) dt$$

$$= x^T \underbrace{\int_{T_0}^{T_f} a a^T dt}_Q x + \underbrace{\int_{T_0}^{T_f} z b^T a^T dt}_{q^T} x + \underbrace{\int_{T_0}^{T_f} b^T b dt}_C$$

x is independent of time

Recall, if $A \in \mathbb{R}^{n \times m}$

$$\int_{T_0}^{T_f} A dt = \begin{bmatrix} \int_{T_0}^{T_f} A_{n1} dt & \dots & \int_{T_0}^{T_f} A_{nm} dt \\ \vdots & \ddots & \vdots \\ \int_{T_0}^{T_f} A_{m1} dt & & \int_{T_0}^{T_f} A_{mm} dt \end{bmatrix}$$

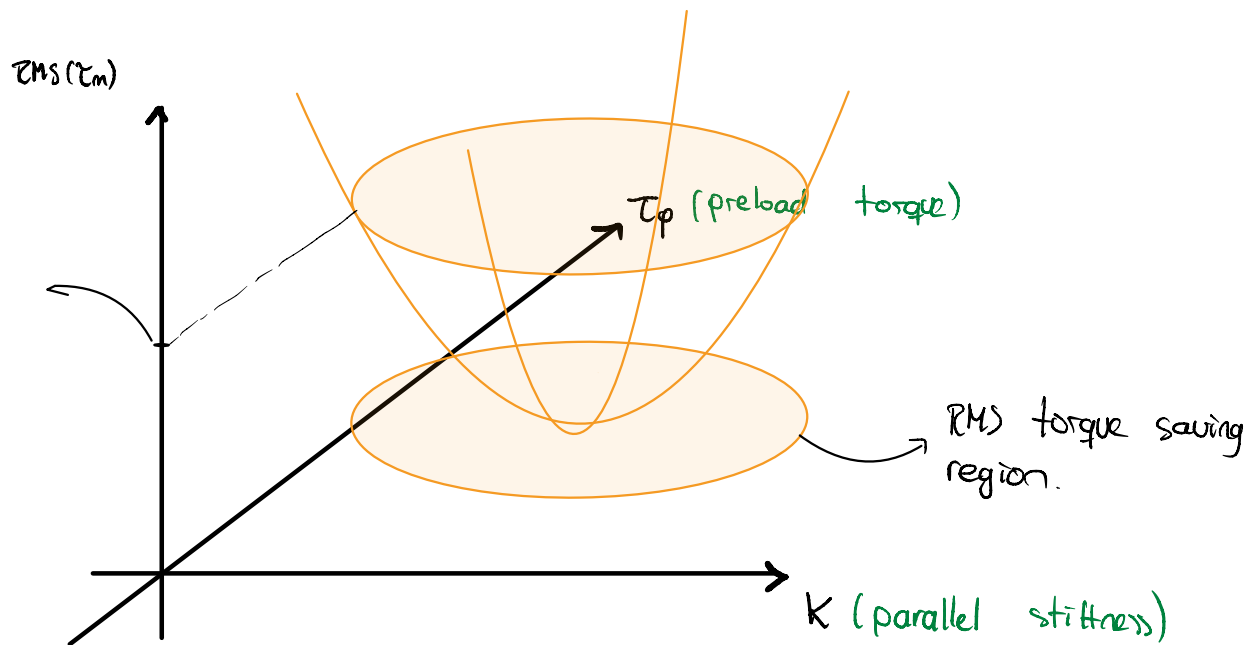
Thus,

$$\int_{T_0}^{T_f} \tau_m^2 dt = x^T Q x + z q^T x + C$$

Positive semi-definite. It is a convex quadratic function.

Intuition,

Typical value of
torque without
parallel spring



Terminology and a quick note
on the transmission of mechanical power.

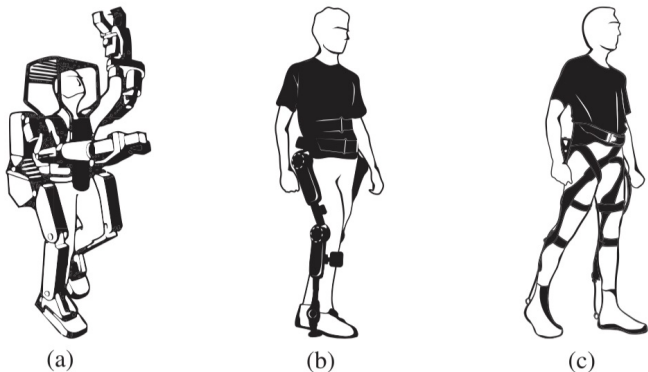


Fig. 1. From the Hardiman to soft robotic suits. (a) Concept of the Hardiman, the exoskeleton proposed by Mosher [3], in 1967, with the goal of augmenting human strength. The device relied on a structural metal frame to transfer loads to the ground and weighed 348 kg. Image adapted from [3]. (b) HAL lower limb exoskeleton (Cyberdyne), 2005. The autonomous exoskeleton weighs 15 kg. Image adapted from [12]. (c) Harvard exosuit, first published in 2013 [13], weighs 15 kg in its latest implementation [14]. Image adapted from [15].

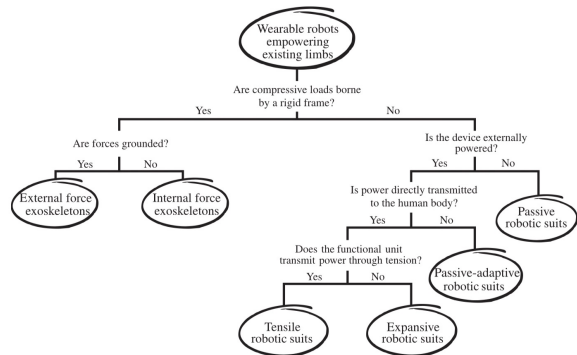


Fig. 2. Taxonomy of soft robotic suits. The classification starts very broadly from wearable robots that assist existing limbs, as defined in [4]. The first dichotomy identifies the difference between exoskeletons and soft robotic suits, the latter not having a rigid frame that bears compressing forces instead of the human skeleton. Soft robotic suits are further categorized according to their source of power. Passive devices exploit gravitational energy, or kinematic energy from the human body, to selectively accumulate and release forces. Passive-adaptive devices exploit an external source of power to modulate their mechanical properties, thus having more versatility over the timing and magnitude of energy accumulation and release. Finally, robotic suits that source power from an external supply and directly deliver it to the human body are divided in those that do so through a tensile element and those that do so through an expansive functional unit.

Soft Robotic Suits: State of the Art, Core Technologies, and Open Challenges

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Driving Robotic Exoskeletons Using Cable-Based Transmissions: A Qualitative Analysis and Overview

Wearable robotics is a field receiving increasing attention from the scientific community. It has great potential to improve rehabilitation process or increase the human capabilities but faces a number of challenges. On the one side, powerful actuation is required, leading to considerable system weight. On the other side, due to the close physical interaction with a human and taking into consideration safety requirements, the displacement of the actuators is crucial to the operational efficiency and functionality of exoskeleton devices. One possible solution for the design of an operational and efficient wearable device is to relocate its actuators out of joints and transmit the force by means of cable-based transmission systems. This paper presents an overview of various cable-based configurations correlated to conventional mechanical designs and their implementation in exoskeleton's structures and an overview of exoskeleton robots including comparison and trend analyses. [DOI: 10.1115/1.4042399]

1 Introduction

Exoskeletons are wearable robotic systems that operate in parallel to the human body. They are defined as devices that can be worn by a human to enhance strength or to restore mobility [1]. Although the first exoskeletons were intended for power augmentation in military applications, such as the Hardiman exoskeleton [2], nowadays, the range of applications has been considerably extended. Several exoskeletons classifications were proposed in literature. They can be classified by general functional purpose: rehabilitation after injury [3,4], daily life assistance [5,6], power

augmentation [7,8], haptic [9–14], and exoskeletons for teleoperation [11,15–17].

Furthermore, exoskeletons might be categorized a function of the human extremity matching in parallel its kinematics chain [18]. Commonly, there can be distinguished upper-limb [19,20] and lower-limb [21] exoskeletons.

Active exoskeletons are often heavy, with a limited power, torque, and energy efficiency, hindering the natural wearer's movements [22]. Often, design of the actuators must be improved to achieve a qualitative operation of the devices [23]. With regard to the wearability of the device, the actual mass and its distribution are factors to be considered.

It is known that adding mass to a human increases the metabolic cost of walking [24,25]. According to biomechanical studies [26,27], adding load at the foot level while walking increases the

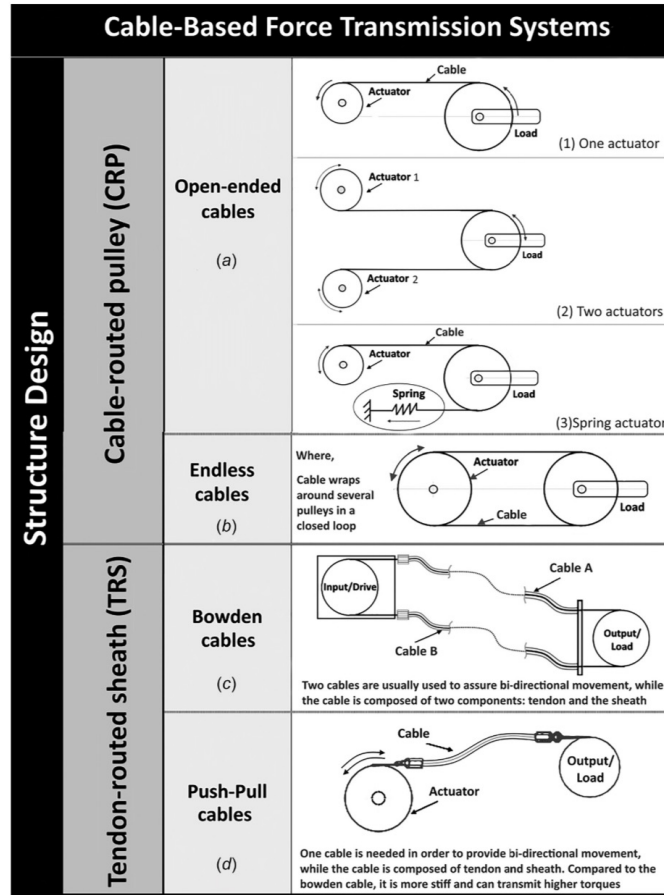


Fig. 1 Conceptual structure designs for cable-based force transmission systems can be classified into two categories: (1) CRP structure, possible in various configurations: (a) open-ended cable configuration and (b) endless cable configuration; (2) TRS structure, where two types can be distinguished: (c) Bowden cable mechanism and (d) push-pull cable mechanism

Mechanical
 losses in power
 transmission

$$P = Tw$$

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